

Class XII | Unit 5 | weight 15%

Database fundamentals

Board weight about 15% | 20 theory periods and 10 lab periods

A database stores related data so that many programs can use it without keeping their own conflicting copies. This lecture moves from that idea to tables, relationships, normal forms and SQL, and ends with the design of a library.

What you should be able to do

- Contrast a flat file with a database management system and name DBMS components, characteristics, models and users.
- Describe the three levels of data abstraction and the basic relational terms.
- Draw entities, attributes and relationships, and apply 1NF, 2NF and 3NF.
- Use keys, integrity constraints and the four relationship types.
- Write SQL for selection, projection, conditions, joins and simple DML.
- Design a library schema through the standard design steps.

Path through the lecture

- 1 Start with one messy spreadsheet of library issues and show the repeated names.
- 2 Normalise that sheet to 3NF on the board. The SQL afterwards uses those tables.
- 3 Keep selection and projection in one pair of examples so the words stay attached to WHERE and to the column list.

5.1 From files to a database

A **flat file** is a single file of records, often one fact repeated on many rows, with no enforced link to other files. A class list in one spreadsheet and an issue list that copies every student's name into every row is a flat-file habit. A **database management system** is software that stores related data, controls access, and lets programs ask questions without each program knowing the physical layout.

Databases run college records, bank accounts, hospital patients, shop stock and examination results. Anywhere the same fact is used by more than one process, a shared database avoids a pile of disagreeing copies.

Components of a DBMS

Component	Meaning
Hardware	The computers and disks the data lives on.
Software	The DBMS itself, together with the application programs.
Data	The stored facts, and the data dictionary that describes them.
Procedures	The rules for backup, access and recovery.
Data access language	The language used to define and manipulate the data. SQL is the one in this course.

Characteristics you should be able to name

Characteristic	What it means here
Controlled redundancy	A fact is stored once and referred to, rather than copied everywhere.
Data accuracy	The stored values match the real world they describe.

Characteristic	What it means here
Integrity	The data obeys rules, such as a mark being within range and a book issue pointing at a real book.
Normalisation	A design discipline that splits tables so that redundancy and update problems shrink.
Security	Only authorised users perform the actions allowed to them.
Scalability	The system can grow in data and users.

The **entity-relationship model** describes the world as entities, attributes and relationships, before tables are chosen. The curriculum calls this the entity relational model. The **relational model** stores data in tables. An **object-relational model** keeps tables and adds object features such as richer data types. Draw the ER model when the question says conceptual design. Draw tables when it says relational.

A **database administrator** controls the structure, the accounts, the backup and the security. A **database application programmer** writes the programs that use the database, such as the library screens.

5.2 Architecture, tables and normal forms

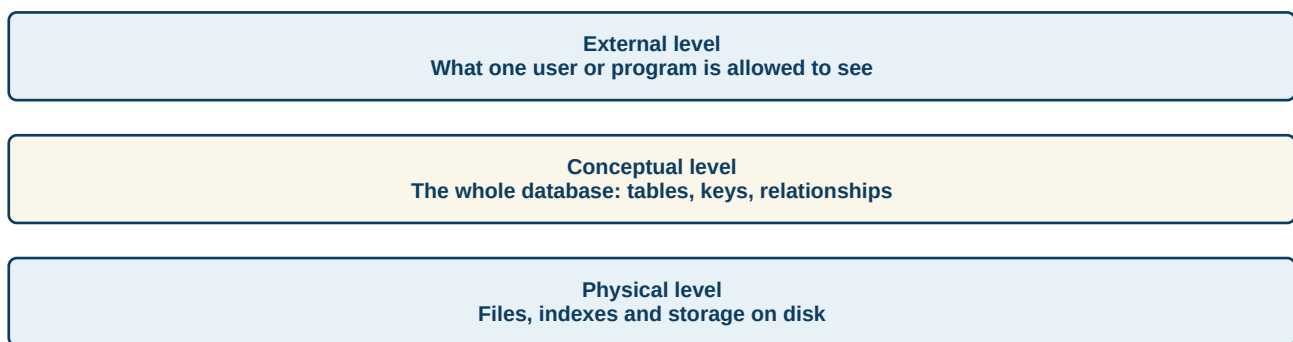


Figure 5.1 Three levels of data abstraction.

The **external level** is a view for one kind of user. A student may see his own issues and not every member's phone number. The **conceptual level** is the whole logical database: all the tables, keys and relationships. The **physical level** is how the data is stored in files and indexes. Users work at the top. The DBMS hides the bottom. This separation is data abstraction.

Relational words

Word	Meaning
Table	A named collection of rows of the same kind. Also called a relation.
Record or tuple	One row.
Field, column or attribute	One named property, such as Title.
Schema	The design of the tables, columns and keys, as distinct from the rows currently stored.
Primary key	A column, or a group of columns, that identifies one row and never repeats.
Foreign key	A column that refers to the primary key of another table.

In an ER diagram an **entity** is a rectangle, an **attribute** is an oval joined to its entity, and a **relationship** is a diamond joined to the entities it connects. Underline the key attribute. For the library, BOOK, MEMBER and the relationship ISSUES are enough to start.

Relationship types

Type	Library meaning
One to one	A member has one current library card. That card belongs to one member.
One to many	One member may borrow many books. From the book side, one copy is issued to one member at a time.
Many to one	Many issue-rows point at one member. This is the same fact as one-to-many, read from the other side.
Many to many	Over time many members borrow many books. A junction table, ISSUE, holds one row per loan and breaks the many-to-many into two one-to-many links.

A normalisation you can reproduce

Start with this repeated sheet. The intended key of one result row is StudentID together with SubjectID.

Before the split

StudentID	SubjectID	StudentName	Dept	HOD	SubjectName	Marks
S1	M	Ayesha	Science	Khan	Maths	80
S1	P	Ayesha	Science	Khan	Physics	74
S2	M	Bilal	Science	Khan	Maths	61

First normal form requires atomic values and no repeating group. Each cell above already holds one value, and each subject is its own row, so the sheet is in 1NF. A cell that contained 'Maths, Physics' would have to be split into two rows first.

Second normal form requires 1NF, and every non-key column must depend on the whole key, not on part of it. StudentName depends only on StudentID. SubjectName depends only on SubjectID. Those are partial dependencies. Move them out. Marks depends on the whole key, so it stays.

After 2NF

Table	Columns	Key
STUDENT	StudentID, StudentName, Dept, HOD	StudentID
SUBJECT	SubjectID, SubjectName	SubjectID
RESULT	StudentID, SubjectID, Marks	StudentID + SubjectID

Third normal form requires 2NF, and no non-key column may depend on another non-key column. In STUDENT, HOD depends on Dept, and Dept depends on StudentID. That is a transitive dependency. Move the department out. The curriculum's heading once writes 3NF beside the letters 2NF. Write **3NF** and mean third normal form.

After 3NF

Table	Columns	Key
DEPT	Dept, HOD	Dept
STUDENT	StudentID, StudentName, Dept	StudentID, and Dept refers to DEPT
SUBJECT	SubjectID, SubjectName	SubjectID
RESULT	StudentID, SubjectID, Marks	StudentID + SubjectID

Changing the head of Science now changes one row in DEPT, not every student row. That is the practical gain.

Integrity constraints protect that design. A **NOT NULL** constraint refuses an empty student name. A **primary key** refuses a duplicate StudentID and refuses to leave it empty. A **foreign key** refuses a RESULT row whose StudentID is not in STUDENT, and refuses a STUDENT whose Dept is not in DEPT.

5.3 SQL

SQL. Structured Query Language. The language used to define tables and to retrieve and change the data in a relational database.

Projection chooses columns. **Selection** chooses rows. In SQL the column list after SELECT is the projection, and the WHERE clause is the selection.

Projection and selection

```
SELECT StudentName, Dept
FROM STUDENT;
```

```
SELECT StudentName, Dept
FROM STUDENT
WHERE Dept = 'Science';
```

The first statement projects two columns of every row. The second also selects the Science rows. Arithmetic in the select list calculates a value: SELECT Marks, Marks + 5 FROM RESULT. Comparison operators are =, <>, >, <, >= and <=. Logical operators are AND, OR and NOT.

An **equijoin** combines rows that have equal key values. A **cross join** combines every row of one table with every row of another and is rarely the result you want. The explicit join and the older WHERE form both appear in classrooms. Know one of them fluently and recognise the other.

Equijoin of students with their marks

```
SELECT STUDENT.StudentName, RESULT.SubjectID, RESULT.Marks
FROM STUDENT
INNER JOIN RESULT
ON STUDENT.StudentID = RESULT.StudentID
WHERE RESULT.Marks >= 50;
```

Create, insert, update and delete

```
CREATE TABLE STUDENT (
    StudentID VARCHAR(6) PRIMARY KEY,
    StudentName VARCHAR(40) NOT NULL,
    Dept VARCHAR(20)
);

INSERT INTO STUDENT (StudentID, StudentName, Dept)
VALUES ('S1', 'Ayesha', 'Science');

UPDATE STUDENT
SET Dept = 'Arts'
WHERE StudentID = 'S1';

DELETE FROM STUDENT
WHERE StudentID = 'S1';

DROP TABLE STUDENT;
```

Microsoft Access uses the same SELECT, WHERE, INSERT, UPDATE and DELETE ideas. In Access query criteria the wildcard for a pattern is often *, while standard SQL uses % with LIKE. Use the form your lab software accepts, and name the operation correctly in the written paper.

5.4 Designing the library

- 1 Purpose: record books, members and which member has which book.
- 2 Tables: BOOK, MEMBER and ISSUE. ISSUE is needed because borrowing is many-to-many over time.

- 3 Fields: BOOK has BookID, Title, Author and Subject. MEMBER has MemberID, MemberName and Class. ISSUE has IssueID, BookID, MemberID, IssueDate and ReturnDate.
- 4 Unique fields: BookID, MemberID and IssueID are the primary keys.
- 5 Relationships: ISSUE.BookID refers to BOOK. ISSUE.MemberID refers to MEMBER. Each relationship is many issues to one book or one member, and the design is in 3NF because names and titles are not repeated on every loan.
- 6 Refine: a ReturnDate may be empty while the book is out, so that column allows null. BookID on a loan must not be null.

Exam point

If the long question is the library, draw three rectangles, two relationships, the primary keys underlined, and the foreign keys marked FK. Then write one SELECT that lists the member name and the book title for loans not yet returned.

Key terms

Term	Meaning to write in an answer
DBMS	Software that manages a shared, structured collection of related data.
Schema	The design of the database.
Primary key	The column or columns that identify one row.
Foreign key	A column that refers to a primary key in another table.
1NF	Atomic values, with no repeating group.
2NF	1NF, and no partial dependency on a composite key.
3NF	2NF, and no transitive dependency.
Selection	Choosing rows, written with WHERE.
Projection	Choosing columns, written in the SELECT list.

Bring it together

- A DBMS shares related data with control of redundancy, integrity and security.
- External, conceptual and physical levels separate the user's view from storage.
- ER diagrams use rectangles, ovals and diamonds. Tables then receive keys.
- 1NF, 2NF and 3NF remove repeating groups, partial dependencies and transitive dependencies.
- SQL selects, projects, joins, inserts, updates and deletes.

Do this in the practical

- 1 Create the STUDENT table, then alter it by adding a column, and drop a practice table you no longer need.
- 2 Insert, update and delete rows.
- 3 Write SELECT statements that filter with WHERE, compare numbers, and join STUDENT to RESULT.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. A primary key:

- A. Identifies one row and does not repeat
- B. Must be a picture
- C. Is always a foreign key of the same table only
- D. Replaces the DBMS

2. Third normal form removes:

- A. Transitive dependencies
- B. All tables
- C. The need for a key
- D. Hardware

3. WHERE in SQL performs:

- A. Selection of rows
- B. Projection only, with no row test
- C. A flowchart
- D. Inheritance

4. A foreign key:

- A. Refers to a primary key, usually of another table
- B. Is a monitor port
- C. Deletes the schema by itself
- D. Is a wireless cell

5. The external level of a database is:

- A. A user's view of part of the data
- B. The physical placement on disk
- C. A C++ destructor
- D. A bus topology

6. INSERT is used to:

- A. Add a row
- B. Draw an entity oval
- C. Boot the operating system
- D. Measure a nibble

7. Many-to-many in a library loan is stored with:

- A. A junction table such as ISSUE
- B. A single cell listing every book
- C. No table at all
- D. A pointer only

8. NOT NULL means:

- A. The column must contain a value
- B. The table is secret
- C. The column is a flowchart
- D. The row is a topology

Answer key

Q	Answer	Why this is the answer
1	A	The primary key uniquely identifies a tuple.
2	A	3NF removes non-key columns that depend on other non-key columns.
3	A	WHERE filters rows. The column list projects columns.
4	A	The foreign key implements a relationship.
5	A	The external level is the view level.
6	A	INSERT is the DML statement that adds data.
7	A	A junction table holds one row per loan and carries both foreign keys.
8	A	NOT NULL is an integrity constraint that rejects an empty value.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Differentiate a flat file and a database.

Model answer

A flat file stores records in one file, and the same fact is often copied onto many rows. A database, managed by a DBMS, stores related tables, reduces that repetition, enforces keys and lets several programs share one controlled copy. A change to a student's name can be made once in the student table.

Q2. Define primary key and foreign key.

Model answer

A primary key is a column or group of columns that identifies one row and cannot be duplicated or left empty. A foreign key is a column that stores the primary key of a row in another table, which creates the relationship. `ISSUE.MemberID` is a foreign key referring to `MEMBER.MemberID`.

Q3. State the rule of each normal form.

Model answer

1NF: each value is atomic and there is no repeating group. 2NF: the table is in 1NF and every non-key column depends on the whole key. 3NF: the table is in 2NF and no non-key column depends on another non-key column.

Q4. Differentiate selection and projection.

Model answer

Selection chooses rows that meet a condition, written with WHERE. Projection chooses columns, written as the list after SELECT. `SELECT StudentName FROM STUDENT WHERE Dept = 'Science'` does both: it projects `StudentName` and selects the Science rows.

Q5. What is an equijoin?

Model answer

An equijoin combines rows from two tables when related columns are equal, typically a foreign key and a primary key. `STUDENT` joined to `RESULT` on `StudentID` produces one row for each result, carrying the student's name. A cross join has no such condition and pairs every row with every row.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain the three levels of abstraction, ER diagrams and normalisation up to 3NF.

- 1 Describe external, conceptual and physical levels.
- 2 Draw or describe entity, attribute and relationship symbols, with BOOK, MEMBER and ISSUES.
- 3 Define 1NF, 2NF and 3NF.
- 4 Walk the student-subject example: partial dependency out at 2NF, HOD out at 3NF.
- 5 Name primary key, foreign key and NOT NULL on the finished tables.

Question 2. Design a library database and write SQL to create a table and to list books on loan.

- 1 State the purpose and the three tables BOOK, MEMBER and ISSUE.
- 2 Give the columns and the primary and foreign keys.
- 3 Justify 3NF: title and member name are not repeated on every loan.
- 4 Write CREATE TABLE for MEMBER or BOOK.
- 5 Write a SELECT with a join that shows member name and book title where ReturnDate is null.